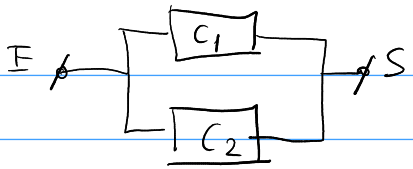


Ex05

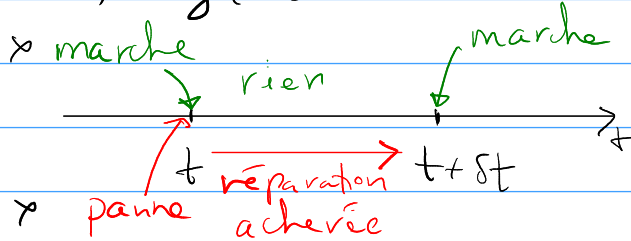


taux de def. $\lambda > 0$

réparation $\mu > 0$

1) $a(t+\delta t) = g(a(t))$

$P(A, B) = P(A|B) \cdot P(B)$



$$a(t+\delta t) = P(\text{Comp en marche à } t+\delta t) \\ = P(\text{Comp en marche à } t+\delta t, \text{Comp en marche à } t) \\ + P(\text{Comp en marche à } t+\delta t, \text{panne à } t)$$

* $P_1 = \frac{P(\text{Comp en marche à } t+\delta t | \text{Comp en marche à } t)}{P(\text{Comp en marche à } t)}$

Loi exp.

sans

mémoire

$$= P(\text{Comp en marche à } \delta t | \text{Comp en marche à } 0) \cdot a(t) \\ = R(\delta t) \cdot a(t) = \underbrace{e^{-\lambda \cdot \delta t}}_{\substack{\text{D.L.} \\ \rightarrow 1 - \lambda \delta t}} \cdot a(t) = (1 - \lambda \delta t) \cdot a(t)$$

* $P_2 = \frac{P(\text{Comp en marche à } t+\delta t | \text{Comp en panne à } t)}{P(\text{Comp en panne à } t)}$

Sans mémoire $\Rightarrow P(\text{Comp en marche à } \delta t | \text{Comp en panne à } 0) = \mu \delta t$

$$= (1 - P(\text{Comp en panne à } t)) \cdot \mu \delta t \\ = (1 - e^{-\mu \delta t}) (1 - a(t)) \approx \mu \delta t (1 - a(t))$$

D.L. $1 - \mu \delta t$

* $a(t+\delta t) = (1 - \lambda \delta t) a(t) + \mu \delta t (1 - a(t))$

2) $a(t+\delta t) = a(t) - \lambda \delta t a(t) + \mu \delta t - \mu \delta t a(t)$

$$e^{a+b} = e^a \cdot e^b$$

$$\lim_{\delta t \rightarrow 0} \frac{a(t+\delta t) - a(t)}{\delta t} = \mu - (\lambda + \mu) \cdot a(t)$$

$$a'(t) = \mu - (\lambda + \mu) \cdot a(t)$$

$$\Leftrightarrow \underbrace{a'(t)} + \underbrace{(\lambda + \mu)a(t)} = \underbrace{\mu}_{\text{ED}_1} \rightarrow \text{ED}_1$$

* Sol homogène: $a'(t) = -(\lambda + \mu)a(t)$
 $\Rightarrow \frac{a'(t)}{a(t)} = -(\lambda + \mu) \Leftrightarrow [\ln(a(t))]'_t = -(\lambda + \mu)$

$\int \Rightarrow \ln(a(t)) = -(\lambda + \mu)t + C$ (const.)
 $\xRightarrow{\text{exp}} a(t) = e^{-(\lambda + \mu)t + C} = \underbrace{e^C}_{K \text{ const}} \cdot e^{-(\lambda + \mu)t}$

$\Rightarrow \boxed{a(t) = K \cdot e^{-(\lambda + \mu)t}}$
 $\hookrightarrow K(t) \Rightarrow \text{variation de constant}$

* Sol non-homogène
 $a(t) = \underbrace{K(t)} \cdot e^{-(\lambda + \mu)t}$

$$a'(t) = K'(t) \cdot e^{-(\lambda + \mu)t} - (\lambda + \mu)K(t) e^{-(\lambda + \mu)t}$$

$$\text{ED}_1 \rightarrow K'(t) \cdot e^{-(\lambda + \mu)t} - (\lambda + \mu)K(t) e^{-(\lambda + \mu)t} = \mu$$

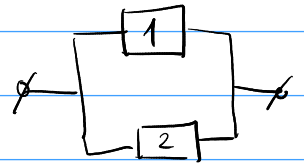
$\Leftrightarrow \underbrace{K'(t) e^{-(\lambda + \mu)t}} = \mu \Leftrightarrow K'(t) = \mu \cdot e^{(\lambda + \mu)t}$
 $\Rightarrow K(t) = \frac{\mu}{\lambda + \mu} \cdot e^{(\lambda + \mu)t} + \underbrace{B}_{\text{const.}}$

$\Rightarrow a(t) = \underbrace{B}_{?} e^{-(\lambda + \mu)t} + \frac{\mu}{\lambda + \mu}$

$$* a(0) = 1 \Rightarrow B \underbrace{e^{-(\lambda+\mu) \cdot 0}}_1 + \frac{\mu}{\lambda+\mu} = 1$$

$$\Rightarrow B = 1 - \frac{\mu}{\lambda+\mu} = \frac{\lambda}{\lambda+\mu}$$

$$\boxed{a(t) = \frac{\lambda}{\lambda+\mu} e^{-(\lambda+\mu)t} + \frac{\mu}{\lambda+\mu}} *$$



$$\begin{aligned} 3) A(t) &= P(\text{sys en marche } \bar{a}t) \\ &= 1 - P(\text{sys en panne } \bar{a}t) = 1 - P(C_1 \text{ et } C_2 \text{ en panne } \bar{a}t) \\ &= 1 - P(C_1 \text{ en panne } \bar{a}t) P(C_2 \text{ en panne } \bar{a}t) \\ &= 1 - (1 - \underbrace{P(C_1 \text{ en marche } \bar{a}t)}) (1 - \underbrace{P(C_2 \text{ en marche } \bar{a}t)}) \\ A(t) &= 1 - (1 - a_1(t)) (1 - a_2(t)) \\ C_1 \text{ et } C_2 \text{ identique} &\Rightarrow a_1(t) = a_2(t) = a(t) \end{aligned}$$

$$\boxed{A(t) = 1 - (1 - a(t))^2} = 1 - (1 - 2a(t) + a^2(t))$$

$$\boxed{A(t) = 2a(t) - a^2(t)}$$

$$A(t) = 1 - \left(\underbrace{1 - \frac{\mu}{\lambda+\mu}}_{\frac{\lambda}{\lambda+\mu}} - \frac{\lambda}{\lambda+\mu} e^{-(\lambda+\mu)t} \right)^2$$

$$A(t) = 1 - \frac{\lambda^2}{(\lambda+\mu)^2} (1 - e^{-(\lambda+\mu)t})^2$$

$$\begin{aligned} a^2 - b^2 &= (a-b)(a+b) \end{aligned}$$

$$A_\infty = \lim_{t \rightarrow \infty} A(t) = 1 - \frac{\lambda^2}{(\lambda+\mu)^2}$$

$$\boxed{a_\infty} = \lim_{t \rightarrow \infty} a(t) = \frac{\mu}{\lambda+\mu} \quad \left[r_\infty = a_\infty \right]_{\mu \rightarrow 0} = 0$$

$$A_\infty = 1 - \frac{\lambda^2}{(\lambda+\mu)^2} = \frac{(\lambda+\mu)^2 - \lambda^2}{(\lambda+\mu)^2} = \frac{\mu(2\lambda+\mu)}{(\lambda+\mu)^2} > \frac{\mu(\lambda+\mu)}{(\lambda+\mu)^2}$$

$$A_{\infty}^{\downarrow} > \frac{\mu}{(1+\mu)} = a_{\infty}^{\downarrow}$$

$$4) M(t) = g(m(t))$$

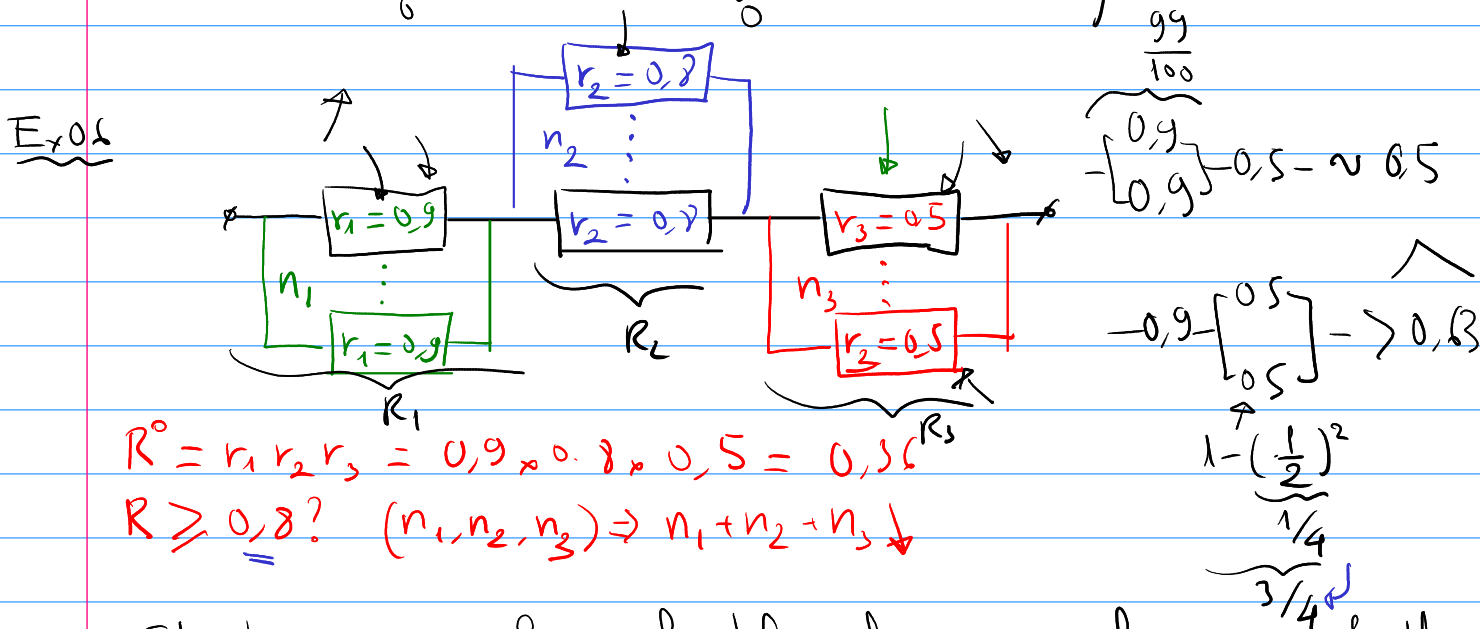
$$M(t) = 1 - \overline{M}(t) = 1 - P(\text{Sys. panne à } t \text{ et n'est pas achevée sur } [0, t])$$

$$= 1 - P(\text{Comp 1 et 2 non réparée sur } [0, t])$$

$$= 1 - (\overline{m}(t))^2 = 1 - e^{-2\mu t}$$

$\downarrow e^{-\mu t}$

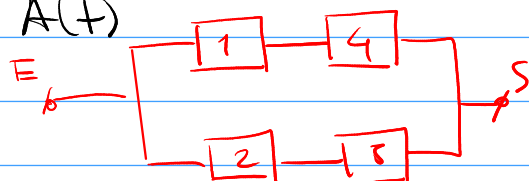
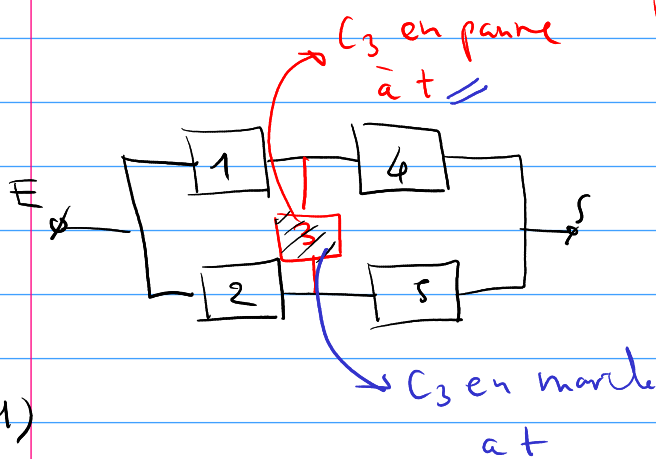
$$MTTR = \int_0^{\infty} \overline{M}(t) dt = \int_0^{\infty} e^{-2\mu t} dt = \frac{1}{2\mu}$$



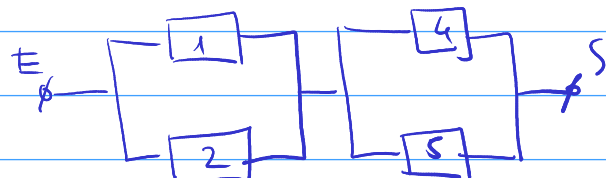
Stratégie : renforcer le bloc des comp's la moins fiable

n_1	n_2	n_3	R_1	R_2	R_3	R	décision
1	1	1	0,9	0,8	0,5	0,36	ajout d'un C3
		2			0,75	0,53	C3
		3			0,875	0,63	C2
						0,756	C3
						0,9375	stop
						0,81	
						> 0,8	

E_{x 07} $a_1(t), a_2(t), \dots, a_5(t), A(t)$



$$A_3(t) = 1 - (1 - a_1 a_4)(1 - a_2 a_5)$$



$$A_{\bar{3}}(t) = (1 - (1 - a_1)(1 - a_2)) \cdot (1 - (1 - a_4)(1 - a_5))$$

$$A_3(t) = P(\text{Sys en marche } \bar{a}t \mid C_3 \text{ en panne } \bar{a}t)$$

$$A_{\bar{3}}(t) = P(\text{Sys en marche } \bar{a}t \mid C_3 \text{ en marche } \bar{a}t)$$

$$P(A|B) = P(A \cap B) / P(B)$$

$$2) A(t) = P(\text{Sys en marche } \bar{a}t)$$

$$= P(\text{Sys en marche } \bar{a}t, C_3 \text{ en panne } \bar{a}t)$$

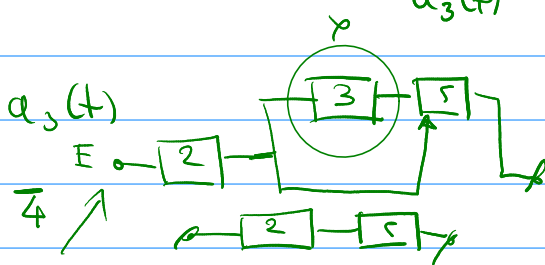
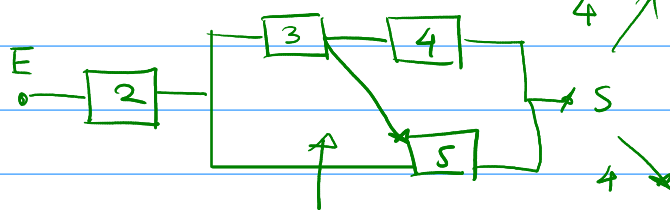
$$+ P(\text{Sys en marche } \bar{a}t, C_3 \text{ en marche } \bar{a}t)$$

$$= P(\text{Sys en marche } \bar{a}t \mid C_3 \text{ en panne } \bar{a}t) \cdot P(C_3 \text{ panne } \bar{a}t)$$

$$+ P(\text{Sys en marche } \bar{a}t \mid C_3 \text{ en marche } \bar{a}t) \cdot P(C_3 \text{ marche } \bar{a}t)$$

$$A(t) = A_3(t) (1 - a_3(t)) + A_{\bar{3}}(t) \cdot a_3(t)$$

C_1 en panne



C_1 en marche